

BANARAS HINDU UNIVERSITY

B.Com. (Hons.) — Semester IV Examination, 2015-16

Paper No. BCH-226: Business Statistics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. All questions carry equal marks (14 marks each).***Question 1.**

What are Index Numbers? How are they constructed? Explain the role of weights in the construction of a general price index number. [14]

OR

The following table gives the average wholesale prices of three commodities for the years from 2011 to 2015. Find Chain Base Index Numbers with 2011 as the base year:

Items	2011	2012	2013	2014	2015
I	2	3	4	5	6
II	8	10	12	15	18
III	4	5	8	10	12

[14]

Question 2.

From the following data, calculate Fisher's Ideal Index Number. Does it satisfy the Factor Reversal and Time Reversal Tests?

Commodity	2014 Base Year		2015 Current Year	
	Price (Rs.)	Quantity	Price (Rs.)	Quantity
A	6	50	10	56
B	2	100	2	120
C	4	60	6	60
D	10	30	12	24
E	8	40	12	36

[14]

OR

An enquiry into the budget of middle class families in Allahabad gave the following information:

Item	Food	Rent	Clothing	Fuel	Miscellaneous
Expenses on	35%	15%	20%	10%	20%
Prices (2009) in Rs.	150	30	75	25	40
Prices (2015) in Rs.	145	30	65	23	45

What changes in cost of living figures of 2015 as compared with that of 2009 are seen? [14]

Question 3.

What do you mean by 'secular trend'? Discuss any two methods of isolating trend values in a time-series and point out their shortcomings. [14]

OR

Below are given the figures of production (in thousand tonnes) of a sugar factory:

Year	2009	2010	2011	2012	2013	2014	2015
Production	77	88	94	85	91	98	90

(a) Fit a straight line by the least squares method and calculate the trend values. [6]

(b) What is the monthly increase in the production of sugar? [4]

(c) Also estimate the production for 2018. [4]

Question 4.

Calculate the seasonal variations using the data given below:

Year	Summer	Monsoon	Autumn	Winter
2011	30	81	62	119
2012	33	104	86	171
2013	42	153	99	221
2014	56	172	129	235
2015	67	201	136	302

[14]

OR

Calculate the seasonal index from the following data using the average method:

Year	Quarterly			
	I	II	III	IV
2011	72	68	80	70
2012	76	70	82	74
2013	74	66	84	80
2014	76	74	84	78
2015	78	74	86	82

[14]

Question 5.

What do you understand by statistical quality control? Discuss its aspects and importance in industry. [14]

OR

Explain the use of p -chart, np -chart and c -chart in this connection for statistical quality control. [14]